

AIM

TA.080 Application Security Architecture

<Company Long Name>

<Subject>

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Introduction

Purpose

The purpose of the **Application Security Architecture** document is to describe the approach being taken to secure access to information and key systems for <Company Long Name> within the scope of the <Project Name>.

It is envisioned that this deliverable will be useful summary of the Application Security Architecture for the following processes in the <Project Name> project:

- Application and Technical Architecture
- Module Design and Build (especially Interface modules)
- Business Requirements Mapping

However, any individual in the <Project Name> project team should have access to the deliverable for reference purposes.

Basic Business Needs

The following areas were covered during the development of this Application Security Architecture:

- a study of the base capabilities of the applications and overall planned environment as they relate to access and information security
- development of the business requirements for access and information security including the need for trusted services for payment, bill presentation and business to business transmittals

Application and Database Security Architecture

Background

There are generally three approaches to provide application data level security in Oracle Applications™. The first and easiest method is to use the data partitioning features that exist in the applications to achieve the data security. For example, General Ledger supports multiple sets of books and data can be secured by set of books in the vanilla application. This then enables legal entities to have their GL data secured from other entities.

The next easiest method is to prevent users from having access to forms, batch programs or reports where they can query and manipulate the secure business objects. This technique provides *effective* data level security by removing access to a complete function in the applications. This is fully supported in Oracle Applications and is achieved by careful definition of menus and responsibilities.

If the standard application level features do not fully meet the requirements, and the technique of removing access to particular application functions cannot be used, it may be necessary to extend the data-level security capability of the applications by customization. A particularly effective technique is the use of secure views in separate Oracle IDs to remove visibility of key business objects by putting a "tag" on the data to indicate which business organization created the data and should have access to it. This "tag" is usually populated by database triggers and may be stored in a descriptive flexfield column on a particular data entity (if possible) to prevent the need for additional table columns on the base package tables.

It is important to understand what is meant by data access in this context and in particular, the distinction between organization and user data access. Data access restrictions arise from the fact that an organization user creates a piece of data within a particular business operating environment and only users in organizations that have an appropriate business operating environment themselves, are able to read and manipulate the data. The data is effectively imbued with access properties from the business operating environment of the creator organization. In order to manipulate a piece of data, a user may be allowed to transform their business operating environment from that of organization A into another environment that does allow access, organization B. Organization A still does not have access to the data *per se*—that privilege resides solely with the user.

Application Security Mapping

The following table lists security requirements by application for each of the implemented Oracle Applications derived from the security policies and the Information Access Model.

The method that will be used to implement the security features are briefly documented in the last column of the table.

Application	Data Class	Detail Function	Standard Oracle	Requirement	Solution
General Ledger	Corporate GL	Corporate on-line/batch transactions and reports	Secure by Set of Books	Only Corporate users should have read/transaction access	Standard Oracle
	Revenue Entity GL	Revenue entity on-line/batch transactions and reports	Secure by Set of Books	Restrict read/transaction access by other entities. Corporate finance users can read/transact any data.	Standard Oracle
Payables	Vouchers	Read/transact access to Revenue Entity Vouchers	Access to all	Only match to POs for Revenue Entity. Corporate finance users can	Access through a secured Oracle ID

Application	Data Class	Detail Function	Standard Oracle	Requirement	Solution
				read/transact any data.	using views/synonyms
		On-line Invoice Approvals	Access to form	Only release holds for Revenue Entity. Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms
	Vouchers, Credit/Debit Memos	Batch Posting of journal entries	Access to Form	Restrict access to vouchers, credit/debit memos by Revenue Entity. Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms
	Checks	Batch Check Printing	Access to all	Restrict access to bank accounts by Revenue Entity. Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms
		Invoice Selection	Paygroups & Priorities	Restrict access by Revenue Entity - probably procedural Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms
Assets	Assets	Run Depreciation	Access to Form	Restrict running by Legal Entity for Legal Entity. Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms

Application	Data Class	Detail Function	Standard Oracle	Requirement	Solution
Receivables	Invoices	AutoInvoice	Access to all	Restrict generation of invoices by Revenue Entity. Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms
	Credit Memos	Apply Credit Memos	Access to all	Restrict invoice selection by Revenue Entity . Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms
	Cash Receipts	Manual Receipts	Access to all	Restrict application to Revenue Entity invoices. Corporate finance users can read/transact any data.	Access through a secured Oracle ID using views/synonyms
	Invoices, Credit Memos	Posting	Access to Form	Restrict posting of invoices and so on by Revenue Entity. Corporate finance users can read/transact any data	Access through a secured Oracle ID using views/synonyms
Manufacturing Applications		Various	Access to all Mfg/Dist Organizations	Restrict transactions for non-management employees to be entered against their Mfg/Dist Organization only. All Mfg/Dist users have read only access to any data.	Modification to login form of Manufacturing Applications to restrict login to Mfg/Dist

Application	Data Class	Detail Function	Standard Oracle	Requirement	Solution
					Organization

Application Security Design

This section describes the designs for functions or features that need to be added to the applications and the overall systems environment to meet the security requirements.

Oracle Applications are developed in adherence to strict design and development standards. These standards insure high quality, portability across multiple platforms, consistent look and feel, ease of maintenance, and compatibility with future versions of Oracle Tools and Application Object Library.

Customizations or extensions to Oracle Applications must also follow strict standards for many of the same reasons, but must also take into consideration factors such as ease of upgrades, preservation of core functionality, maintainability, and the design review process.

Trusted Services

Trusted services involve the use of key encryption schemes, secure-token authentication and other means of validating that the requester and provider of a transaction are the intended parties. Further the transaction may be encapsulated in an encryption scheme.

Payment Authorization Services

These services interact with bank and clearing organization servers to secure pre-authorization for debts.

The project requirements for this type of service are:

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-
-

Payment Transaction Services

These services interact with bank and clearing organization servers to transmit actual payment and remittance information related to particular transactions.

The project requirements for this type of service are:

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-
-

Bill Presentation Services

These services interact directly business to business or business to intermediary to business to present bills or invoices for subsequent payment.

The project requirements for this type of service are:

-
-
-

Business to Business Transmittals

These services interact directly business to business or business to intermediary to business to transmit critical transactions such as purchase orders, advance ship notices, receipt confirmations, and so on.

The project requirements for this type of service are:

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-
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Application Security Configuration

The section describes in general terms how the application will support the security requirements. The application configuration may use standard application features that the application uses to provide the supported security functions, or they may be custom extensions or modifications to the applications that are needed to provide security at the application level above and beyond what is provided in the standards applications.

Database Security Design

This section describes the design of a specific custom database architecture to support the database level security requirements.

Database Security Configuration

The section describes in general terms how the database will be partitioned to support the general security requirements. The partitioning may arise from a standard database level features that the application uses to provide the supported security features, or they may be custom modules that are needed to provide security at the database level above and beyond what the applications support.

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date